MILITARY SPECIFICATION

VALVE MANIFOLDS, HIGH PRESSURE

AIR REDUCING

1. SCOPE

1.1 Scope. - This specification covers the design, construction, test and operating requirements for high pressure air reducing valve manifolds for the ASROC launcher.

1.2 Classification. - Manifolds shall be of the following classes, as specified (see 6.1):

Class I - 4500/100 pounds per square inch gage (psig) manifold.
Class II - 4500/300 psig manifold.

Class I and class II manifolds differ only in capacity and the type of reducing valve employed; therefore, differentiation will be made only in the capacity, regulator and end connection requirements of this specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact) Shipboard Application, Tests for.
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-S-7742 - Screw Threads, Standard, Optimum Selected Series: General Specification for.
MIL-P-15024 - Plates, Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manual, Service (Instruction Book) for Shipboard Electrical and Mechanical Equipment.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).

STANDARDS

MILITARY

MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

DRAWINGS

BUREAU OF SHIPS

810-1385884 - Union Fittings 6000 PSI WOG IPS.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. -

3.1.1 Description. - The manifold assemblies shall be designed to accept up to 5000 psig air pressure and shall be capable of reducing this pressure to a lower pressure, the value of which shall be as specified in table I. The manifold shall perform this function with a component arrangement as shown on figure 1.

3.1.2 Size, weight and reliability. - The manifold shall be as small, compact and light as possible, consistent with satisfactory performance, reliability, durability, strength and quietness. Space, weight and reliability shall be the prime consideration in the evaluation of the equipment for subject application.

3.1.3 Materials. - Materials used in the construction of the manifold shall be of the best commercial grade suitable for the intended use. All materials shall be compatible with ships air containing sea water vapor (100 percent humidity) and 2190TEP lubricating oil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:56Z
Check the source to verify that this is the current version before use.

Table I - Regulation

Normal inlet pressure (psig) range		Reduced pressure at lockup conditions (nominal)	Allowable variation of reduced pressure at lockup conditions throughout the inlet pressure range (psi)	Maximum allowable "droop" of reduced pressure at full flow conditions (psi)
High	Low			
4500	1500	100	± 2	-15
4500	1500	300	± 5	-15

3.1.3.1 Dissimilar metals. - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in direct contact with each other.

3.1.4. Interchangeability. - All parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation and performance.

3.1.5. Safety wiring. - Accidental loosening of all threaded parts shall be prevented by safety wiring, self-locking bolts and nuts, and other approved methods.

3.1.6. Threads. - All threads shall conform to MIL-S-7742.

3.1.7. Mounting. - The body of the manifold shall be designed for permanent installation in the ships pneumatic system. All of the individual components used in the manifold assembly shall be easily accessible for service or removal from the manifold.

3.1.8 Identification plates and markings. - Identification plates and markings shall be furnished in accordance with MIL-P-15024 and MIL-STD-130. In addition, the manifold shall have an identification plate which shows the schematic diagram of the manifold and detailed instructions for its use.

3.1.9 Environmental conditions. - Manifold performance shall not be adversely affected by varying ambient conditions within the following limits:

(a) Temperature - 28°F to 125°F
(b) Pressure - 30 inches ± 6 inches Hg.
(c) Salt spray - Exposure to atmosphere containing salt laden moisture, to meet the applicable requirements of MIL-E-5272.
(d) Vibration - To meet vibration test requirements for type I of MIL-STD-167.
(e) Shock - Manifold shall meet high shock impact requirements as defined in MIL-S-901. If the manifold satisfactorily passes the shock test, it shall be repaired in accordance with the applicable paragraph for disposition of shock tested equipment as specified in MIL-S-901 and offered for acceptance as a production unit.

3.1.10 Noise level. - Design, construction and workmanship shall be such that minimum practical noise level will result under all operating conditions specified herein.

3.2 Detail requirements. -

3.2.1 Design and test pressures. - Design and test pressures of the manifold shall be as shown in table II.

3.2.2 Ports. - Manifold ports for classes I and II shall be equipped with 6000 psig unions in accordance with Drawing 810-1385884. The thread piece may be secured to the manifold body with a straight thread and "O" ring seal. Sizes shall be as shown in table III.

3.2.3 Flow direction. - The manifold shall be designed for in to out flow only.

3.2.4 Capacity. -

3.2.4.1 Class I manifold shall be designed for a maximum capacity of 250 SCFM with an inlet pressure of 1500 psig and the lock up pressure set at 100 psig.

3.2.4.2 Class II manifold shall be designed for a maximum capacity of 20 SCFM with an inlet pressure of 1500 psig and the lock up pressure set at 300 psig.

3.2.4.3 Flow limiting orifice. - The manifolds shall incorporate a built-in orifice of a size

Source: https://assist.dla.mil -- Downloaded: 20260-19T15:56Z
Check the source to verify that this is the current version before use.

sufficient to allow the valves to meet all flow conditions, but which will limit the capacity to 25 percent above the maximum specified capacity.

3.2.5 Leakage. - Leakage shall be as follows:

(a) External - none
(b) Internal - none when tested in accordance with 4.2.

3.2.6 Filter. - The manifold shall contain a filter capable of 5 micron particles removal. The rated capacity of the filter shall be the same or greater than that of the regulator with a maximum pressure drop of 15 psi.

3.2.7 Differential pressure indicator. - The manifold shall contain a differential pressure indicator which will give a visual indication when the pressure differential across the filter exceeds 35 psi.

3.2.8 Hand valves. - Hand valves indicated on figure 1 shall be of the poppet type and shall be of a design to limit diametrical clearance between the valve poppet and seat to a maximum of 0.002 inches during the first two turns of the valve handwheel. Operating torques shall not exceed 40 inch-pounds. These valves shall incorporate a soft seating feature.

Table II - Design and test pressures

Class	Manifold inlet and outlet		Regulater dome when protected with a relief valve	
	Proof	Burst	Proof	Burst
I	7500	15000	200	400
II	7500	15000	450	900

Table III - End connection sizes

Class	Inlet, IPS	Outlet, IPS
I	Inch 3/8	Inches 1-1/4
II	3/8	3/8

3.2.9 Regulators. - Regulators shall be designed as follows:

(a) The regulators shall be designed to meet the capacity requirements specified in 3.2.4 and shall operate without hunting, chattering, or excessive noise, or vibration, when subjected to all operating conditions.
(b) Regulators shall be capable of withstanding the sudden application of maximum inlet pressure to the reducer such as may be induced by a solenoid valve opening without exceeding the lock up limits of pressures specified in table I.
(c) Regulators for class I manifolds shall be pilot operated. Specifically, the regulators shall be self-loading, and shall operate in such a way that a sudden application of upstream pressure shall result in a gradual, surge-free pressure at the outlet port. The regulators shall be constructed so that when the system is vented (upstream and downstream), the poppet shall be closed.
(d) Regulators for class II manifolds shall be either directly operated or pilot operated.
(e) The pilot regulators shall incorporate a built-in relief valve to protect the dome of the main valve from over pressurization.
(f) The regulators shall close automatically upon failure of the main actuating piston, seal, spring, or diaphragm.
(g) The pressure regulators shall be of a "Balanced Design" which will automatically compensate for wide variations in upstream pressure. Table I indicates the allowable fluctuation of the downstream pressure as the upstream pressure varies within the specified limits.

3.2.10 Gage. - The manifolds shall include a 2-1/2 inch pressure gage which will indicate the

Source: https://assist.dla.mil -- Downloaded: 202640-19T15:56Z
Check the source to verify that this is the currentversion before use.

regulated pressure. The gage on the class I manifold shall have a full scale reading of 200 psig. The gage on the class II manifold shall have a full scale reading of 400 psig. Gages for class I and class II manifolds shall have an accuracy of plus or minus 3 percent over the entire range span. Gage connection shall be made with a straight male thread and "O" ring seal.

3.3 Repair parts and special tools. - Repair parts and special tools shall be furnished in accordance with MIL-P-15137. Special tools are defined as those tools not listed in the Navy Stock List of General Stores. (Copies of this stock list may be consulted in the office of the Government inspector.) Quantities on a per ship basis shall be as shown in table IV.

Table IV - Repair parts and special tools

Item	Quantity per ship
Springs	25 percent but not less than one set
Diaphragms Gaskets, "O" rings and seats	100 percent
Poppets and seat Retainers	25 percent but not less than one set
Special tools for assembly and disassembly	One set

3.4 Manuals. - Manuals shall be furnished in accordance with MIL-M-15071.

3.4.1 Insert requirements. - Insert requirements shall be as follows:

(a) Inserts may be the equivalent of the manufacturer's standard commercial instruction and parts lists; however, they shall not contain information on other equipment. The inserts shall contain the basic principles of operation of the equipment, and sufficient information for its installation, adjustment, and maintenance. In addition, a schematic diagram of the manifold should be given with detail instructions for its use.
(b) A list of repair parts shall be included.
(c) Reduced size prints of drawings specified in 3.5 shall be included.
(d) A table of gasket and seals shall be included listing all applicable dimensions and other data necessary for identification and reordering.
(e) Charts showing volume, SCFM versus differential pressure for each class of manifold shall be included in the manual.

3.5 Drawings. - Class A and B drawings shall be furnished in accordance with MIL-D-963.

3.5.1 Drawings required shall include an outline, assembly and sub-assemblies as necessary to clearly depict the design and construction of the equipment. The foregoing procedure does not preclude the possibility that a print of a detail drawing may be necessary in some particular instance to clarify a design point. Prints of detail drawings shall also be provided for all replaceable subassemblies or parts specified by "repair parts list" except where parts can be adequately specified or are in common commercial use and can be referenced to commercial standards.

3.5.2 Preliminary drawings. - Four blueprint copies of preliminary drawings shall be forwarded with the quotation. These drawings should supply sufficient information to insure a complete engineering evaluation of the design. (The preliminary drawings are to include dimensions (including servicing spaces) piping connections and sizes, location or mounts, an estimated dry weight and performance characteristics.) These drawings are necessary for preliminary machinery and piping arrangements as well as bid evaluation.

3.5.3 Drawing list. - Within four weeks from date of contract, the vendor shall submit a list of all working drawings showing vendor number, master plan number and titles with estimated submittal dates.

3.5.4 Existing master drawings. - Where previously approved and validated master drawings are applicable under the contract, four blueprint copies shall be submitted to the bureau or agency concerned for approval.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:56Z
Check the source to verify that this is the current version before use.

3.5.5 Within 90 days from date of contract, four blueprint copies of all new class A and class B drawings in accordance with MIL-D-963 shall be submitted to the bureau or agency concerned for approval. Certification data sheets are required as a supplement to all master drawings. Where existing certification data (except contract numbers and so forth) are applicable to an identical component on a later ship, extension of the data shall be accomplished by adding additional hull numbers, new purchase order, contract number, and so forth, and the revision block of the bureau or agency number shall be revised to next highest letter. Notes of exception will not be used. Where deviations occur, new certification data will be required. After approval and validation by the bureau or agency concerned, one clear, sharp and legible cloth tracing of all new class A and B drawing certification data sheets shall be furnished in first class condition.

3.6 Workmanship. - The manifolds shall be free from defects which may effect their appearance and which may effect their operation. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First unit tests. - A complete agenda of the test procedures shall be submitted by the contractor to the Bureau of Ships for approval prior to starting the tests. After approval of the test procedures, one manifold of each class shall be tested to determine conformance with the following:

(a) Vibration (see 3.1.9(d))
(b) Shock - (see 3.1.9(e))
(c) Design and test pressure - (see 3.2.1)
(d) Capacity - (see 3.2.4)
(e) Leakage - (see 3.2.5 and 4.2.1)
(f) Hand valves - (see 3.2.8)
(g) Regulators - (see 3.2.9(g)).

4.2.1 Leakage limitations. - Manifolds of each class shall be subjected to a lockup condition leakage test as follows: With 4500 psi at the inlet, and less than 6 cubic inches of volume at the outlet port, the outlet gage shall be monitored for a period of 10 minutes during which time there shall be no visible evidence of regulated pressure.

4.2.2 After initial approval, the first unit tests need not be repeated, on subsequent orders. A test report on the initial tests is acceptable proof of satisfactory performance and the report should be submitted to the bureau or agency concerned. No changes or deviations from the first unit tested will be acceptable without the approval of the bureau or agency concerned.

4.3 Quality control inspection. - A complete agenda of the examination and test procedures shall be submitted by the contractor to the Bureau of Ships for approval prior to starting the examination and tests. After approval of the procedures, each manifold shall be examined and tested in accordance with the approved procedures to determine conformance to the requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing. - Manifolds shall be prepared for delivery in accordance with the manufacturer's best commercial practice for domestic shipment—immediate use which will insure acceptance and safe delivery at destination. Manifolds shall be cleaned by process C-1 of MIL-P-116. All ports and openings shall be plugged or sealed with approved corrosion-resistant material to prevent entrance of moisture and dirt.

5.2 Repair parts and tools. -

5.2.1 Packaging, packing and marking. - Onboard repair parts and tools shall be packaged and packed for shipment in a manner which will insure acceptance and safe delivery at destination. Onboard repair parts and tools shall be marked for domestic shipment.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class required (see 1.2).

Notice - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity
Navy - Ships
(Project 4730-N047Sh)

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:56Z
Check the source to verify that this is the currentversion before use.

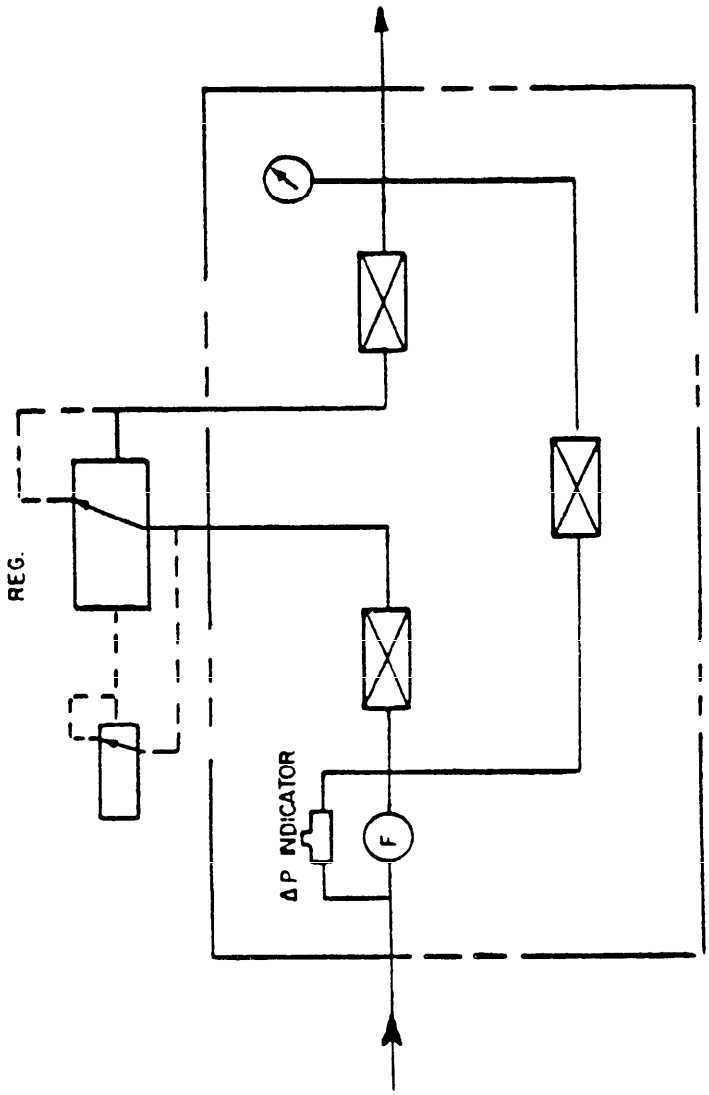


Figure 1—Arrangement

SH-6076

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:56Z
Check the source to verify that this is the current version before use.

S[illegible]FICATION ANALYSIS SHEET
[illegible] PS-4853 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships.

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? IF THE ANSWER IS "YES", IN WHAT WAY?

☐ YES ☐ NO

4. REMARKS (Attach any pertinent data which may be of use in improving this specification.) PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY (Print Name and activity)	DATE

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:56Z
Check the source to verify that this is the current version before use.

Fold

DEPARTMENT OF THE NAVY
BUREAU OF SHIPS
WASHINGTON 25, D. C.

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
NAVY DEPARTMENT

CHIEF, BUREAU OF SHIPS
SPECIFICATIONS AND STANDARDIZATION BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON 25, D. C.

Fold

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:56Z
Check the source to verify that this is the current version before use.